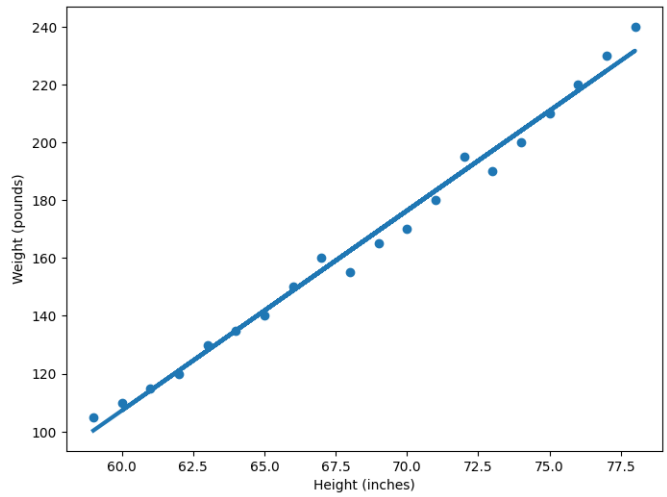
Assignment #10

1. Here is a hypothetical dataset called `height_weight.csv`. (a) Plot the scatter diagram and draw the regression line for y on x.



(b) What is the R^2? What does this mean?

.988. It is the fraction of variability in y explained by the regression model.

2. A small data set is shown below.

x	y
1	2
2	3
3	1
4	5
5	6

The fitted regression line is $y = x + .4$.

(a) What happens to the regression line if you add 3 to each value of y instead of interchanging the columns.

$y = x + 3.4$. The y-intercept is increased by 3, but the slope stays the same.

(b) What happens to the regression line if you double each value in x.

The slope is halved. $y = .5x + .4$.

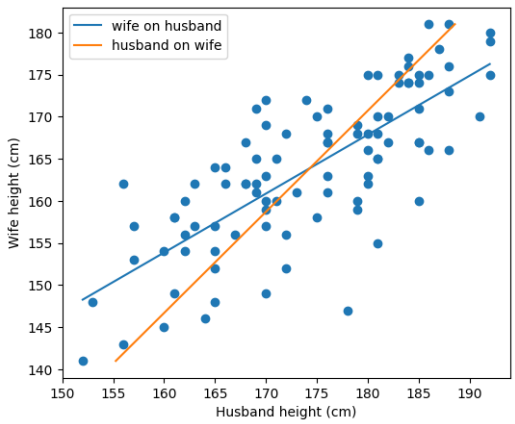
3. The data file `couplesheights.txt` has been provided with the height of the husband in the first column in cm and the height of the wife in the second column in cm.

   (a) Plot a scatter diagram.

   (b) Plot the fitted regression line of the height of the wife as a function of the height of the husband.

   (c) Now plot the fitted regression line of the height of the husband as a function of the height of the wife. Clearly label each line.

   (d) Are the two regression lines the same or different? Why?



The two regression lines are different, because one regression line picks off the centers of the vertical strips and the other picks off the center of the horizontal strips.

4. The regression method predicted that a student at the 90th percentile on the SAT would only be at the 69th percentile on first-year GPA.

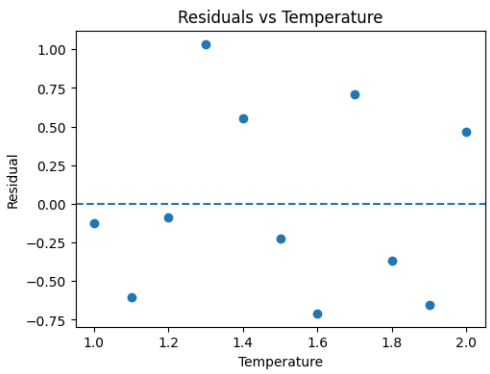
   (a) True or false, and explain: a student at the 69th percentile on first-year GPA should be at the 90th percentile on the SAT. (Hint: Look at the previous question)

   False. The regression lines are different as shown from the previous questions.

   (b) Someone who scored at the 30th percentile on the SAT on average should score _____ the 30th percentile on his/her first-year GPA.
   Options: lower than, about equal to, higher than
   Why?

higher than. Regression to the mean.

5. A study was made on the amount of converted sugar in a certain process at various temperatures. The data is given in the file, `temperature_sugar.txt` with the temperature in the first column and the converted sugar in the second column.
(a) Estimate the linear regression line.
(b) Estimate the mean amount of converted sugar produced when the coded temperature is 1.75.
(c) Plot the residuals versus temperature. Comment.

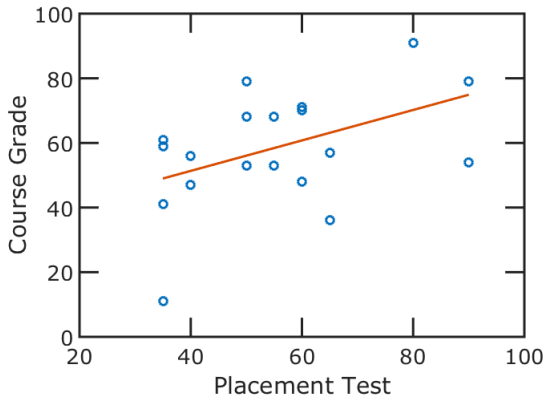
(a) $y = 1.8x + 6.4$
(b) 9.6
(c) 

Residuals appear to be random as desired.

6. For the previous question, (a) evaluate s^2; (b) construct a 95% confidence interval for β_0; (c) construct a 95% confidence interval for β_1.

(a) $s^2 = .4$ (b) 6.4136 ± 2.0917 (c) 1.809 ± 1.36

7. A mathematics placement test is given to all entering freshmen at a small college. A student who receives a grade below 35 is denied admission to the regular mathematics course and placed in a remedial class. The placement test scores and the final grades for 20 students who took the regular course were recorded. The data is given in file, `placement_grade.txt` with the placement test scores in the first column and the course grade scores in the second column.
(a) Plot a scatter diagram.
(b) Find the equation of the regression line to predict course grades from placement test scores.
(c) Graph the line on the scatter diagram.
(d) If 60 is the minimum passing grade, below which placement test score should students in the future be denied admission to this course?



(a)
(b) y = .471 x + 32.5
(d) 58.4

8. Test the hypothesis that $\beta_0 = 10$ in the previous problem against the alternative that $\beta_0 < 10$. Use a 0.05 level of significance.

$H_0 : \beta_0 = 10$
$H_0 : \beta_0 < 10$
$t = 1.78$
$P = .95$
Fail to reject H_0

9. Six different machines are being considered for use in manufacturing rubber seals. The machines are being compared with respect to tensile strength of the product. A random sample of four seals from each machine is used to determine whether the mean tensile strength varies from machine to machine. The following are the tensile-strength measurements in kilograms per square centimeter $\times 10^{-1}$:

1	2	3	4	5	6
17.5	16.4	20.3	14.6	17.5	18.3
16.9	19.2	15.7	16.7	19.2	16.2
15.8	17.7	17.8	20.8	16.5	17.5
18.6	15.4	18.9	18.9	20.5	20.1

Perform the analysis of variance at the 0.05 level of significance and indicate whether or not the mean tensile strengths differ significantly for the six machines.

$H_0 : \mu_1 = \mu_2 = \ldots \mu_6$
H_1 : At least two of the means are not equal
Use the f-test.
$P = .90$ Do not reject H_0

10. The data in the following table represent the number of hours of relief provided by five different brands of headache tablets administered to 25 subjects experiencing fevers of 38° C or more. Perform the analysis of variance and test the hypothesis at the 0.05 level of significance that the mean number of hours of relief provided by the tablets is the same for all five brands. Discuss the results.

A	B	C	D	E
5.2	9.1	3.2	2.4	7.1
4.7	7.1	5.8	3.4	6.6
8.1	8.2	2.2	4.1	9.3
6.2	6	3.1	1	4.2
3	9.1	7.2	4	7.6

$H_0 : \mu_1 = \mu_2 = \ldots \mu_5$
H_1 : At least two of the means are not equal
Use the f-test.
$P = .0015$
Reject H_0